Artificial Intelligence (AI-2002)

Lecture 3: Intelligent Agents

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Agents and Environments

An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.

Agents and Environments

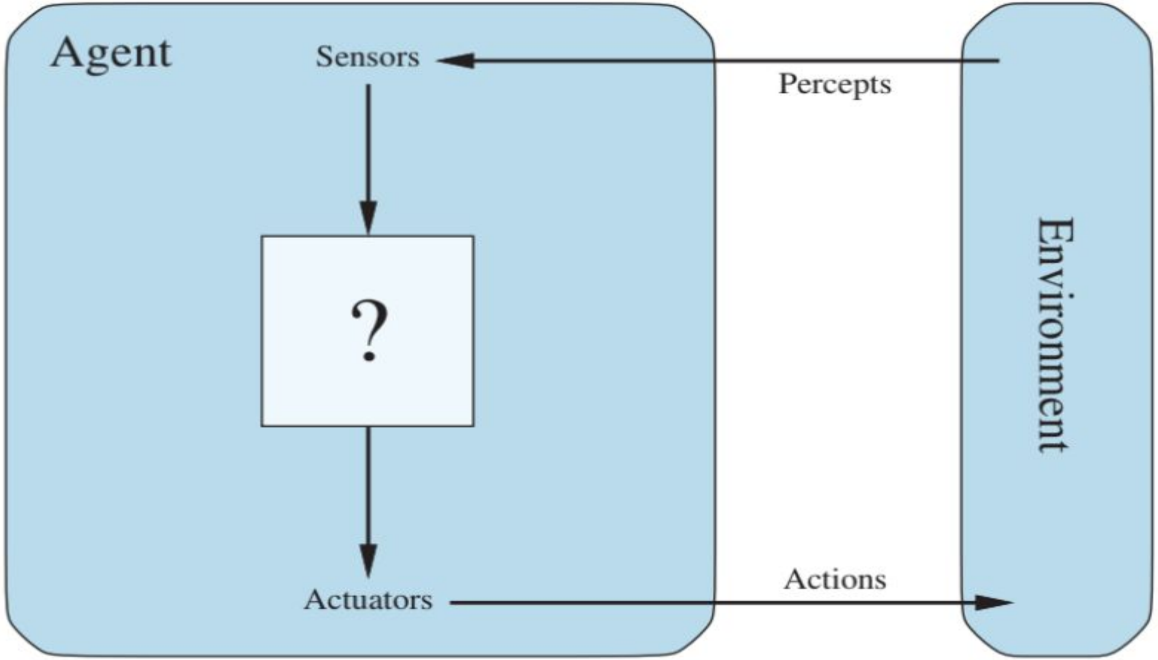

Figure 2.1 Agents interact with environments through sensors and actuators.

Agents and Environments

- Percept refers to the content perceived by an agent's sensors at any given moment.
- Percept Sequence represents the complete history of everything the agent has perceived.
- An agent's choice of action is based on its built-in knowledge and the entire percept sequence observed so far.

Agents and Environments

Agent Function:

- Describes the agent’s behavior mathematically.
- Maps any given percept sequence to a corresponding action.

Vacuum-Cleaner World:

- Vacuum-Cleaner World: A robotic agent operates in a two-square environment (A and B), where each square can be dirty or clean. The agent perceives its location and dirt status.
- Actions: Move left, move right, suck up dirt, or do nothing. The agent starts in square A.
- Simple Agent Function: If the current square is dirty, suck; otherwise, move to the other square.

Vacuum-Cleaner World:

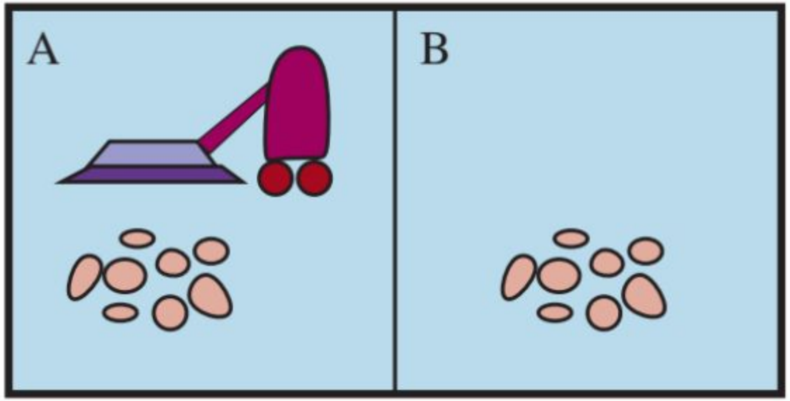

Figure 2.2 A vacuum-cleaner world with just two locations. Each location can be clean or dirty, and the agent can move left or right and can clean the square that it occupies. Different versions of the vacuum world allow for different rules about what the agent can perceive, whether its actions always succeed, and so on.

Vacuum-Cleaner World:

Percept sequence	Action
$[A, Clean]$	*Right*
$[A, Dirty]$	*Suck*
$[B, Clean]$	*Left*
$[B, Dirty]$	*Suck*
$[A, Clean], [A, Clean]$	*Right*
$[A, Clean], [A, Dirty]$	*Suck*
$\vdots$	$\vdots$
$[A, Clean], [A, Clean], [A, Clean]$	*Right*
$[A, Clean], [A, Clean], [A, Dirty]$	*Suck*
$\vdots$	$\vdots$

What makes an agent good or bad, intelligent or stupid?

Good Behavior: The Concept of Rationality

- A rational agent is one that does the right thing. Obviously, doing the right thing is better Rational agent than doing the wrong thing, but what does it mean to do the right thing?

Performance measures

- Moral Philosophy offers various ideas about what constitutes the "right thing."
- AI typically follows consequentialism, evaluating behavior based on the consequences of actions.
- Agent's Behavior: An agent generates a sequence of actions based on its percepts, causing the environment to change states.
- Performance Measure evaluates whether the sequence of states is desirable, determining the agent's performance.

Performance measures

- Creating an effective performance measure can be difficult and may lead to unintended behaviors.
- Measuring performance by the amount of dirt cleaned could lead the agent to repeatedly clean and dump dirt to maximize its score.
- Reward the agent for maintaining a clean floor (e.g., awarding points for clean squares at each time step and penalizing for electricity usage and noise).
- Design performance measures based on the desired outcomes in the environment, rather than prescribing specific agent behaviors.

Rationality

What is rational at any given time depends on four things:

- The performance measure that defines the criterion of success.
- The agent's prior knowledge of the environment.
- The actions that the agent can perform.
- The agent's percept sequence to date.

Rationality

This leads to a definition of a rational agent:

- For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Rationality

- Consider the simple vacuum-cleaner agent that cleans a square if it is dirty and moves to the other square if not.
- Is this a rational agent? That depends!
- First, we need to say what the performance measure is, what is known about the environment, and what sensors and actuators the agent has.

Rationality

Let us assume the following:

- The performance measure awards one point for each clean square at each time step, over a “lifetime” of 1000 time steps.
- The “geography” of the environment is known a priori but the dirt distribution and the initial location of the agent are not.
- The only available actions are Right, Left, and Suck.
- The agent correctly perceives its location and whether that location contains dirt.

Rationality

- Under these circumstances the agent is indeed rational; its expected performance is at least as good as any other agent's.

Rationality V/S Omniscience

- We need to be careful to distinguish between rationality and omniscience.
- An omniscient agent knows the actual outcome of its actions and can act accordingly; but omniscience is impossible in reality.

Rationality V/S Omniscience

Consider the following example: I am walking along the Champs Elysees one day and I see an old friend across the street. There is no traffic nearby and I'm not otherwise engaged, so, being rational, I start to cross the street. Meanwhile, at 33,000 feet, a cargo door falls off a passing airliner and before I make it to the other side of the street I am flattened. Was I irrational to cross the street? It is unlikely that my obituary would read "Idiot attempts to cross street."

Rationality V/S Omniscience

This example shows that rationality is not the same as perfection. Rationality maximizes expected performance, while perfection maximizes actual performance. Retreating from a requirement of perfection is not just a question of being fair to agents. The point is that if we expect an agent to do what turns out after the fact to be the best action, it will be impossible to design an agent to fulfill this specification—unless we improve the performance of crystal balls or time machines.

Think of an example of a rational agent and define how it is rational.